

Dexter Fernandes

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https://github.com/Dexter-Fernandes

AI Engineer

Applied AI Engineer with 3+ years building production Python systems, now focused on AI agents, RAG, and LLM evaluation. Built a stateful LangGraph agent orchestrating retrieval, correctness, and grounding tools to diagnose RAG failures, with structured Pydantic outputs benchmarked against a deterministic baseline. Designed a dual-track evaluation layer (deterministic checks + LLM-as-judge), exposed through typed MCP tools for programmatic failure diagnosis.

PROJECTS

FinanceBench RAG Evaluation Harness

Jun 2026

- Built a stateful **LangGraph** agent orchestrating retrieval, correctness, and grounding tools through conditional workflows to diagnose RAG failures, with structured Pydantic outputs benchmarked for diagnostic accuracy, latency, and tool usage against a deterministic baseline.
- Built a **hybrid-retrieval RAG** pipeline for financial-document QA (**BM25, dense embeddings, reciprocal rank fusion, cross-encoder reranking**), with three isolable retrieval modes (closed-book, oracle-context, dense) to separate retrieval quality from generation quality.
- Designed a dual-track evaluation layer pairing deterministic checks (exact/normalised match, scale-aware numeric matching) with **LLM-as-judge** scoring, and a retrieval-vs-answer metric join attributing wrong answers to retrieval, generation, or grounding failure.
- Benchmarked the pipeline against **Snowflake Cortex Search** on identical chunks and metrics, exposing runs, results, and **failure diagnostics** through typed **MCP** tools with Markdown/CSV/JSON reporting.

WORK EXPERIENCE

Independent Projects & Upskilling

Sep 2024 - Present

Self-directed

- Built a stateful LangGraph agent orchestrating retrieval, correctness, and grounding tools through conditional workflows to classify RAG failures, benchmarking diagnostic accuracy, latency, and tool usage against a deterministic baseline.
- Completed Nvidia "Building RAG Agents with LLMs" (Jul 2026), covering RAG, agents, and evaluation with LangChain and LangSmith.

Esbaa • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Owned the end-to-end AI pipeline for real-time perception in harsh industrial environments, covering data curation, model training through deployment and on-site validation, improving F1 by 15% while maintaining deterministic 24/7 operation on British Petroleum rigs.
- Applied statistical and experimental methods including Bayesian optimisation, ablation studies, hypothesis testing, and confidence intervals to support model selection, threshold tuning, and failure analysis in production.
- Built multimodal data ingestion, cleansing, training, and validation pipelines across image, video, and sensor data using Python, Pandas, Airflow, MLflow, DVC, Optuna, and CVAT, processing 3.6 million records and reducing model rollout time from 3 days to under 24 hours.
- Presented model performance, failure modes, and deployment trade-offs to engineering teams, client stakeholders, and field operators, turning messy production signals into clear optimisation recommendations.

Lead Software Engineer - AI

- Led a team to design, train, and deploy production object detection models for industrial safety applications, reducing false positives by 33% while meeting latency and power constraints for Austin Industries and John Deere.
- Built automated labelling, dataset versioning, and performance-tracking workflows, repurposing existing detection models for annotation and cutting manual labelling effort by 80% on large real-world datasets.
- Worked with hardware, software, QA, and client teams to translate model performance, latency, and failure analysis into rollout decisions for deployed safety systems.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

Dissertation: Radiological SLAM with LiDAR Odometry

- Dissertation: designed and implemented a Radiological SLAM system on a Clearpath Husky platform, integrating LiDAR and radiological data for localisation, mapping, and inspection workflows with Python.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

- Built a strong foundation in signals, systems, embedded hardware, and control, which later supported work in real-time AI, sensor systems, and robotics software.

CERTIFICATIONS

Building RAG Agents with LLMs

Jul 2026

Nvidia

Deep Learning Specialisation

Jul 2019

deeplearning.ai

SKILLS

- **AI-Assisted Development:** Claude Code, ChatGPT Codex
- **AI Engineering and Evaluation:** LLM-as-judge, RAG evaluation, document-grounded QA, retrieval metrics, evidence matching, citation checks, hallucination detection, Prompt engineering, error analysis, structured outputs, RAG pipelines, dense retrieval, FAISS vector store, ChromaDB, hybrid retrieval, Pydantic v2, FastAPI, MCP, agent orchestration, tool orchestration, stateful agent workflows
- **Experimentation and Reliability:** failure-mode analysis, benchmarking, regression checks, hypothesis testing, ablation studies, post-deployment validation, production monitoring
- **Python and ML Engineering:** Python, Pandas, NumPy, scikit-learn, PyTorch, MLflow, DVC, Apache Airflow, Optuna, LangChain, LangSmith, PyDantic, FastAPI, Snowflake, LangGraph
- **Production Systems:** Docker, CI/CD, Git, structured logging, observability, health checks, Apache Kafka, Linux, schema-validated output contracts, Microsoft Azure, SQL